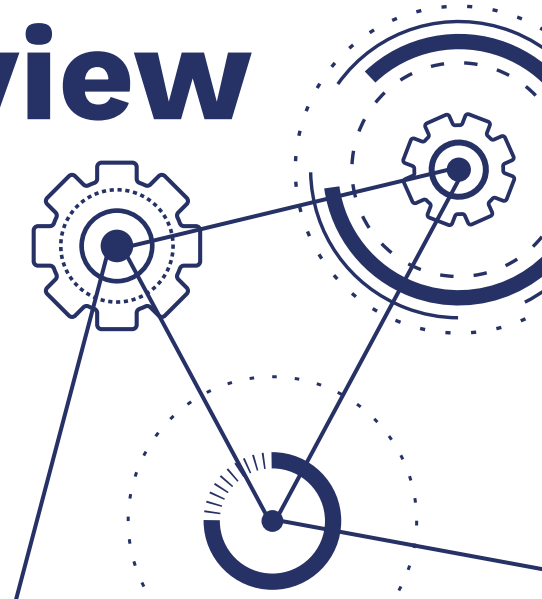




Manufacturing Review

Ola, Elysia, Lindsay, Evan, Miles, Nathan



Criteria

Criteria

Presentation Structure

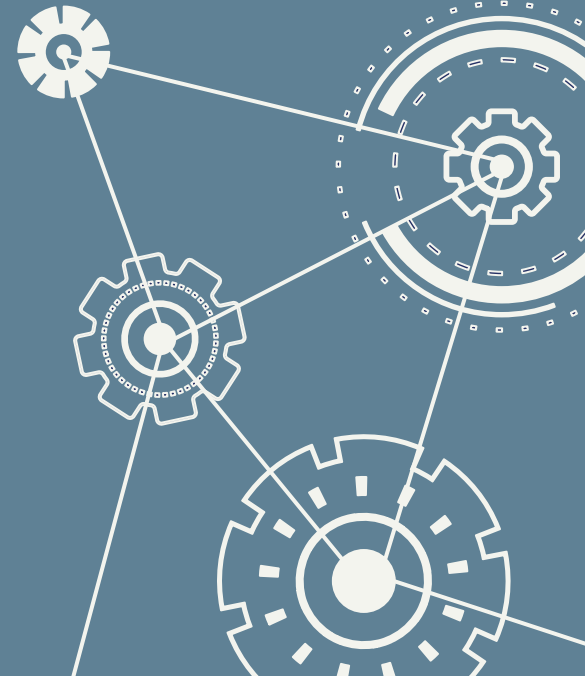
- be ready to present CAD and manufacturing results (quality of your manufactured parts) for all parts of the yo-yo
 - bring example parts and molds
 - be ready to discuss the biggest challenges and roadblocks your team has overcome
 - have a clear outline and description of any remaining issues, including plans for addressing those concerns
- have measurement data (bluetooth caliper measurements) ready for **100 of your body and ring parts (as final as possible)**
- students will be graded on
 - completeness of the required elements
 - clarity of presentation



Design for Manufacturing

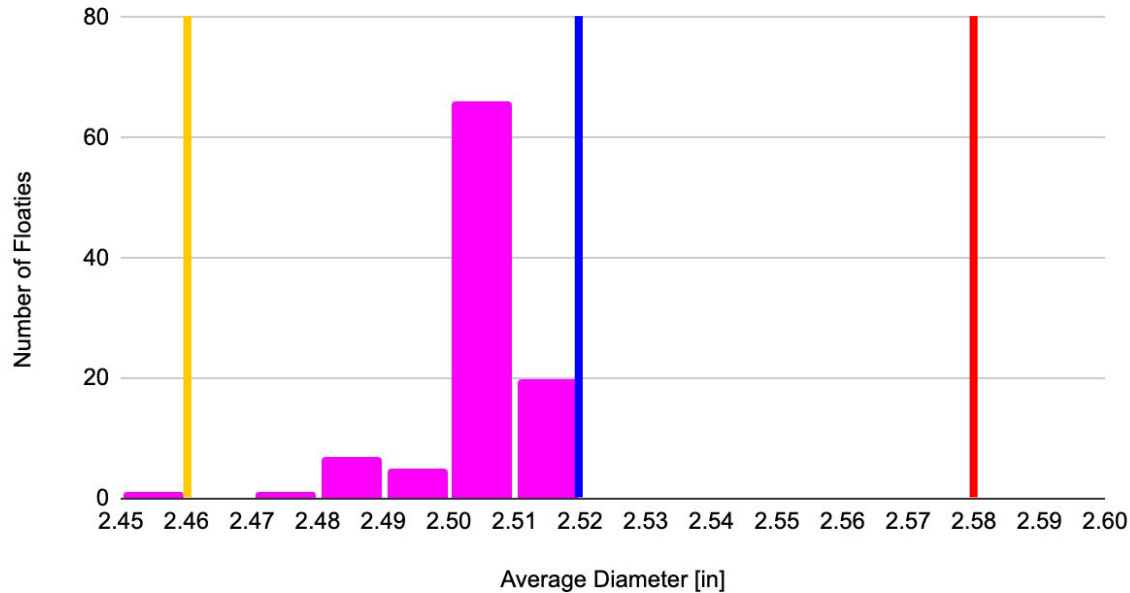
- how did you work towards delivering high quality components/assemblies?
- which features ended up being the hardest to manufacture?
 - how does this experience match up to what you thought was going to be hardest?
- students will be graded on:
 - effectiveness and appropriateness of manufacturing results

Summary of Results



Distributions - Floatie

Kir-B Floatie Press Fit Diameters



$$\mu = 2.4965$$

$$\sigma = 0.015$$

$$n = 100$$

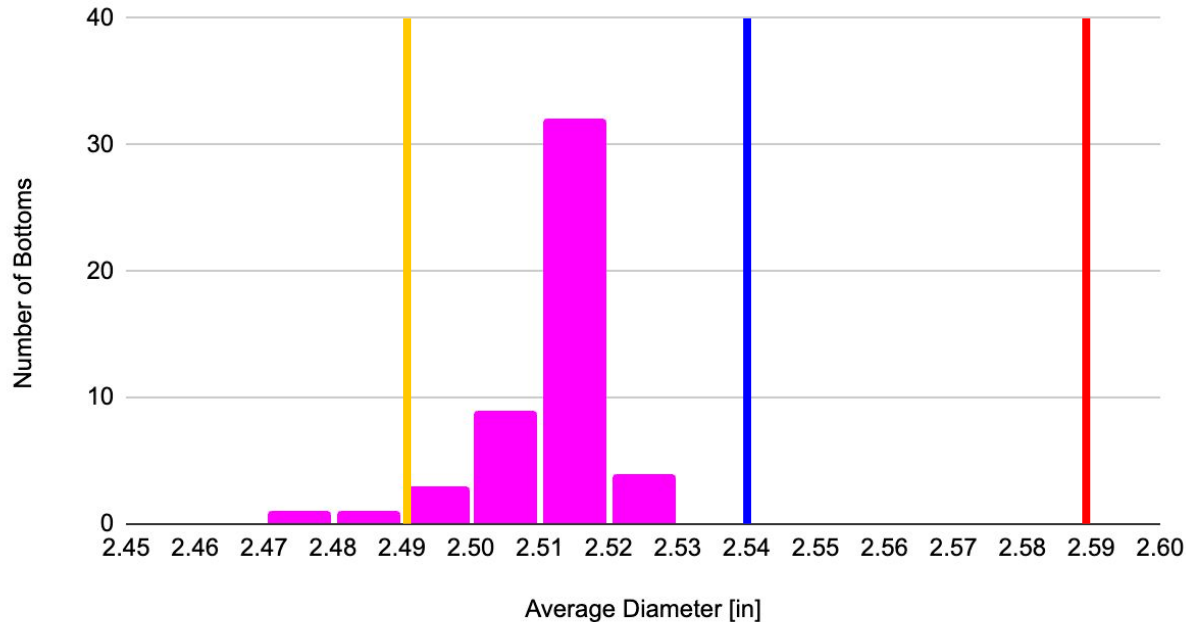
CAD SPEC: 2.52" +/- 0.06

USL = 2.58

LSL = 2.46

Distributions - Lower body

Kir-B Lower Body Press Fit Diameter



$$\mu = 2.508984$$

$$\sigma = 0.008$$

$$n = 50$$

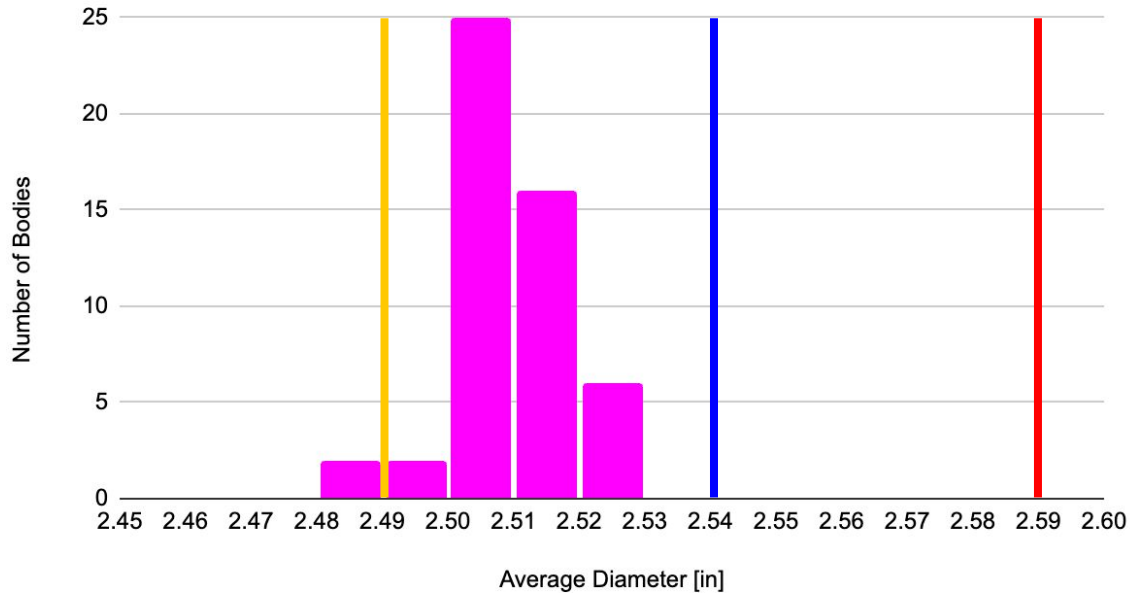
CAD SPEC: 2.54" +/- 0.05

USL = 2.59

LSL = 2.49

Distributions - Upper body

Kir-B Upper Body Press Fit Diameter



$$\mu = 2.5121$$

$$\sigma = 0.009$$

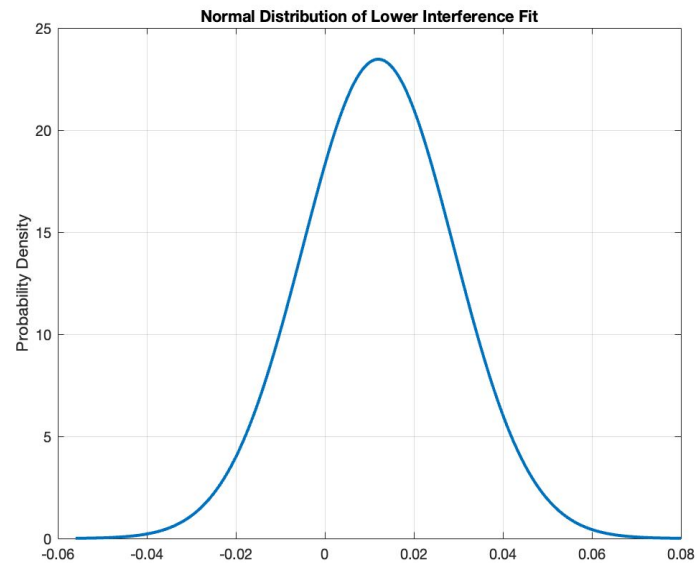
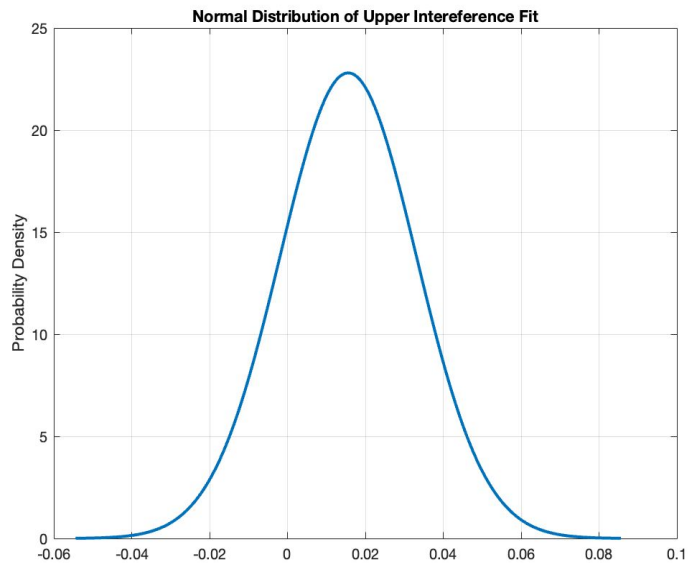
$$n = 50$$

CAD SPEC: 2.54" +/- 0.05

USL = 2.59

LSL = 2.49

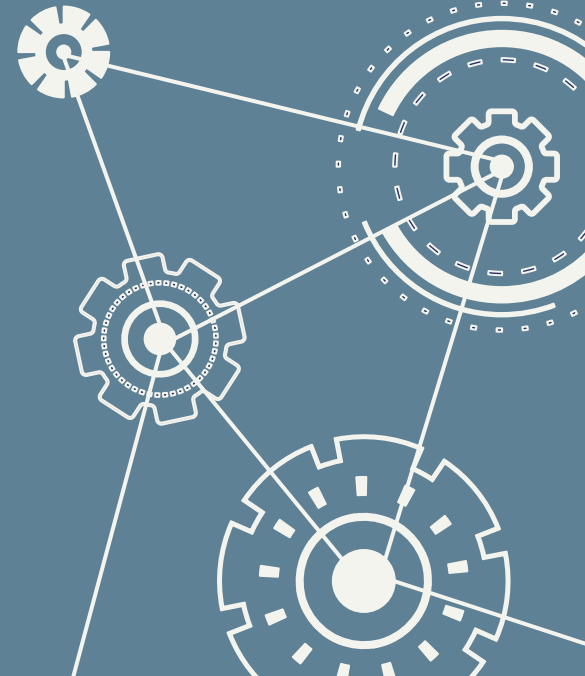
Distributions - Interference



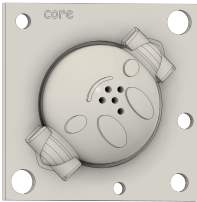
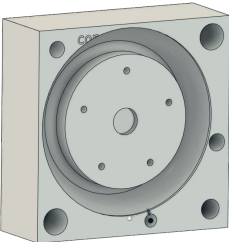
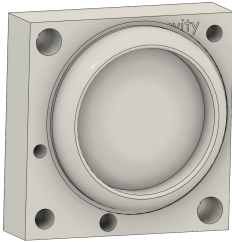
Distributions - Interference

Dimension	Mean (in.)	Standard Deviation (in.)
OD upper body	2.5121	0.009
OD lower body	2.508984	0.008
ID of floatie	2.4965	0.015
Interference (Upper)	0.0156	0.0175
Interference (Lower)	0.012	0.017

Design for Manufacturing



Yo-yo IM part outline *[design review]*

KirB Part (quantity)	Core	Cavity
Upper Body (1)	 A 3D model of the Upper Body Core, showing a square block with a central circular cavity and four corner holes. The word "core" is visible in the top left corner.	 A 3D model of the Upper Body Cavity, showing a square block with a central circular cavity and four corner holes. The word "cavity" is visible in the top right corner.
Lower Body (1)	 A 3D model of the Lower Body Core, showing a square block with a central circular cavity and four corner holes. The word "core" is visible in the top left corner.	 A 3D model of the Lower Body Cavity, showing a square block with a central circular cavity and four corner holes. The word "cavity" is visible in the top right corner.
Floatie [yo-yo body] (2)	 A 3D model of the Floatie Core, showing a square block with a central circular cavity and four corner holes. The word "core" is visible in the top left corner.	 A 3D model of the Floatie Cavity, showing a square block with a central circular cavity and four corner holes. The word "cavity" is visible in the top right corner.

Floatie Evolution

First Attempt

The interference was too high, discrepancies between CAD and manufactured molds
Determining proper IM parameters



Re-measuring

Re-calculating press fit

Finalized floatie



Floatie Evolution

First Attempt



Re-measuring

IM'ed a sample size to accurately measure
resulting diameter
Mold smaller than CAD

Re-calculating press fit

Finalized floatie

Critical Diameter	
CAD	2.542 in
Mold	2.5307 in
IM	2.487 in

Floatie Evolution

First Attempt

Floatie
Body, mold diameter: 2.5307 in
IM part: 2.486 in

$$\text{Actual Shrinkage} \rightarrow 1 - \frac{2.486 \text{ in}}{2.5307 \text{ in}} = 0.0177 = 1.77\%$$

→ New

predicted shrinkage
1.15%

(Bottom)
Ring, mold diameter: 2.5596 in
IM part: 2.527 in

$$\text{Shrinkage} \rightarrow 1 - \frac{2.527 \text{ in}}{2.5596 \text{ in}} = 0.0127 = 1.27\%$$

actual shrinkage

Re-measuring

we want → 0.02" interference!

we have → 2.527 in - 2.486 in = 0.041

2.527 in - 0.02 = 2.507 in for the floatie IM part

Re-calculating press fit

Comparing predicted vs. actual shrinkage

Re-making mold with adjusted diameter

Finalized floatie



2.507 in

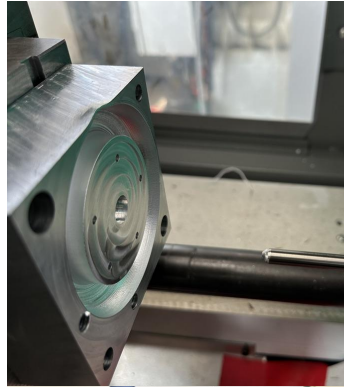
$$\frac{x - 2.507}{x} = 0.0177$$

$$\frac{2.507}{x} = .9823$$

x = 2.552 in
body mold diameter

Floatie Evolution

First Attempt

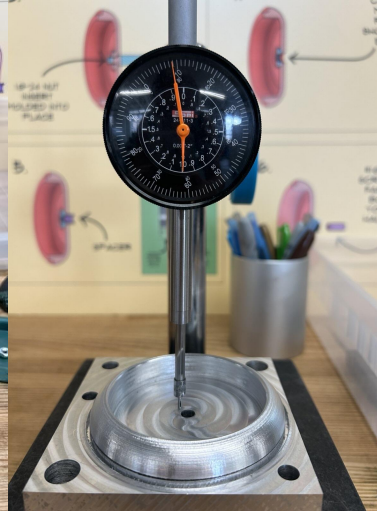


Re-measuring

Re-calculating press fit

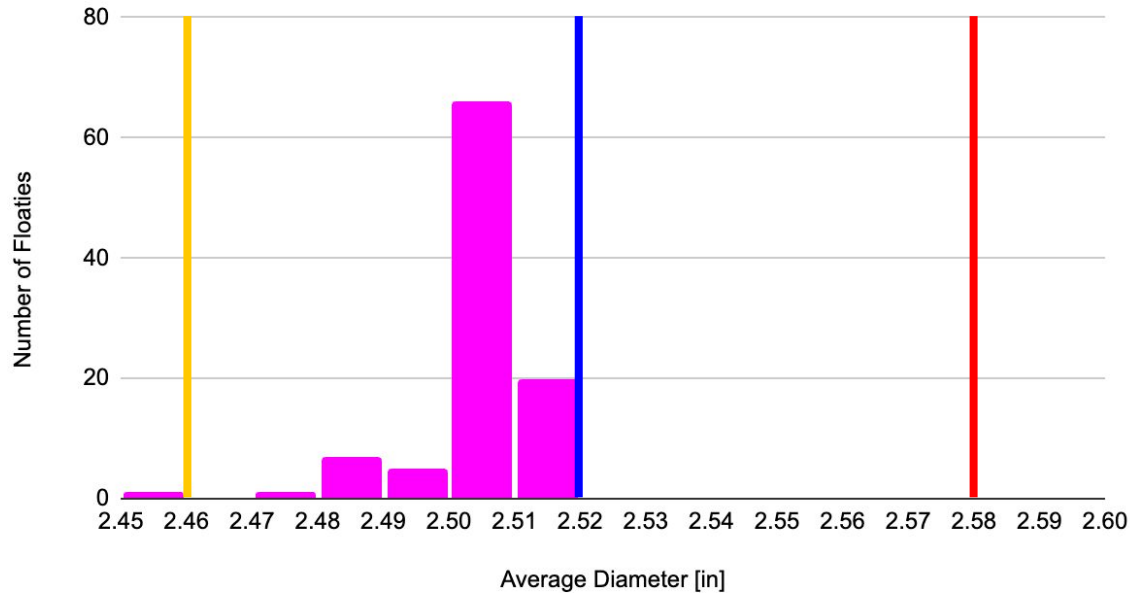
Finalized floatie

Mistakes along the way
Shoulder bolt measurements
Perfect fit



Distributions - Floatie

Kir-B Floatie Press Fit Diameters



$$\mu = 2.4965$$

$$\sigma = 0.015$$

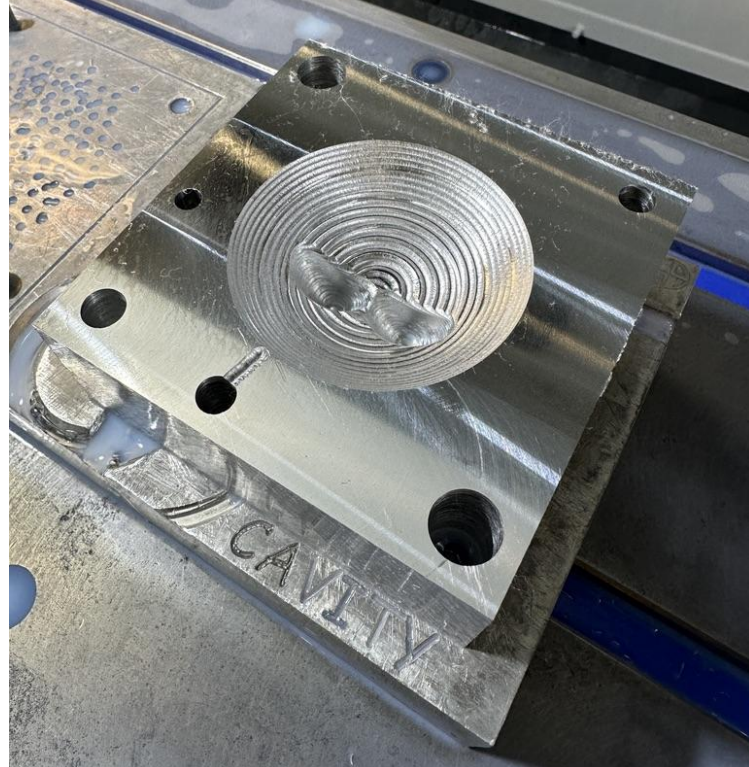
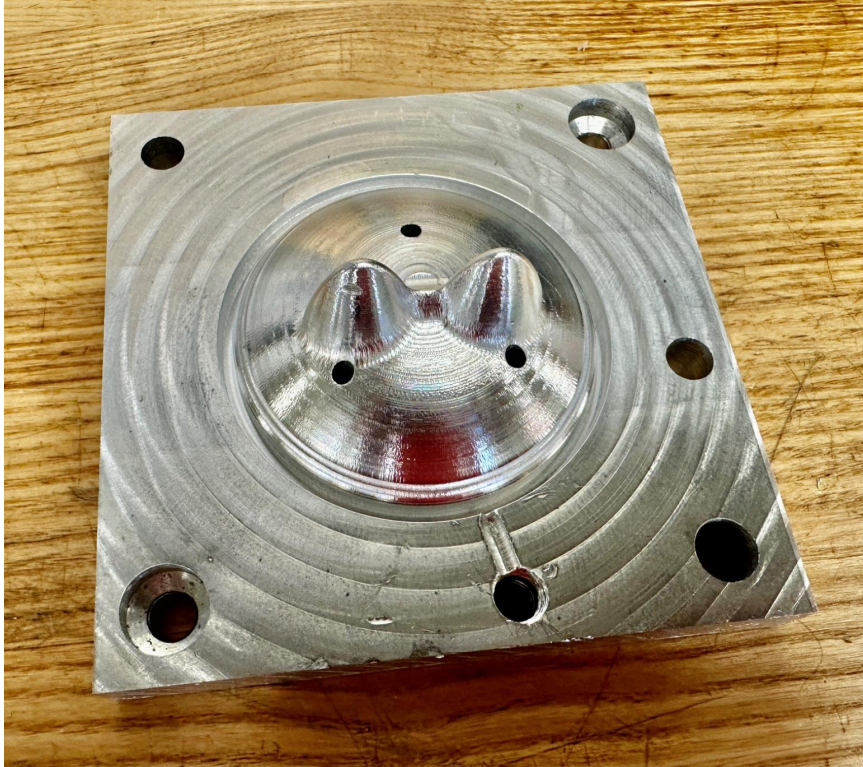
$$n = 100$$

CAD SPEC: 2.52" +/- 0.06

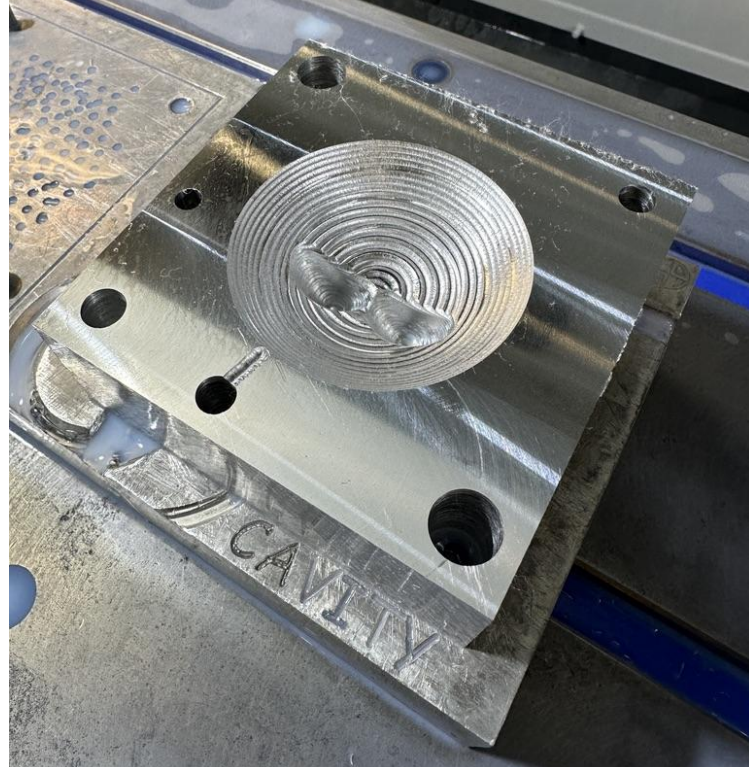
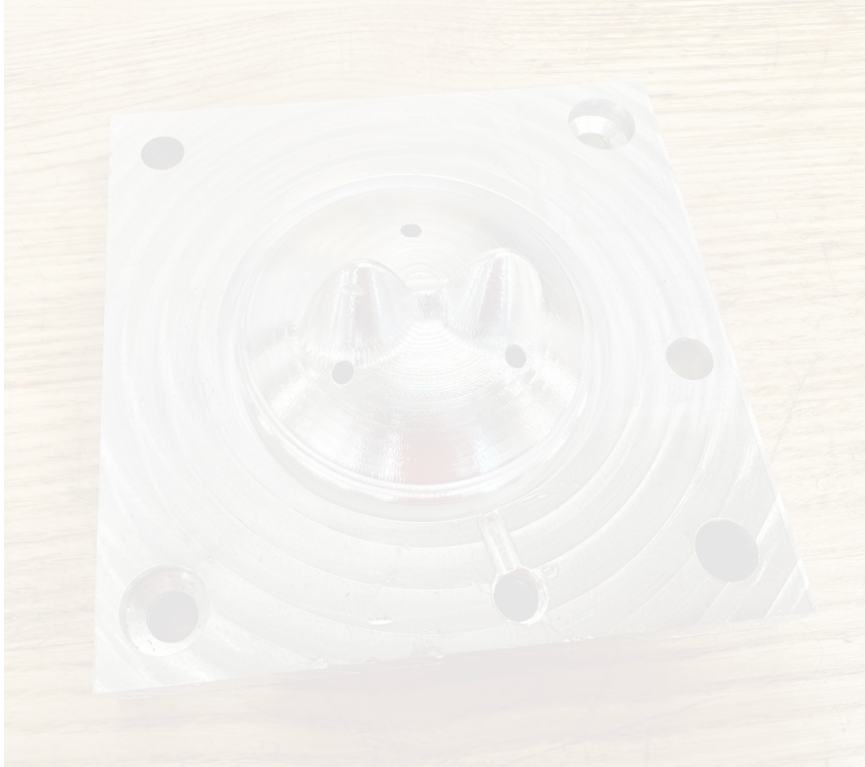
USL = 2.58

LSL = 2.46

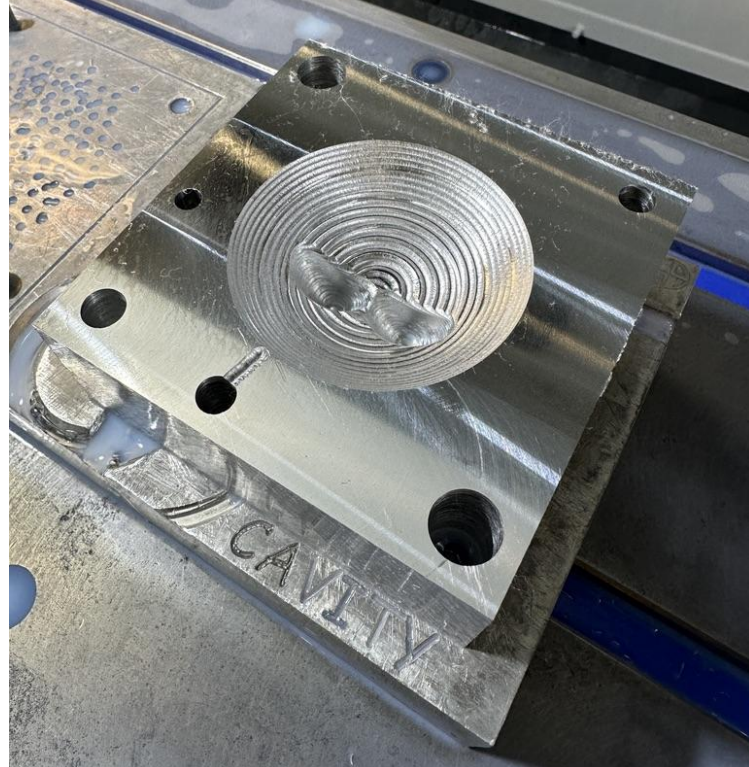
Lower Body Evolution



Lower Body Evolution



Lower Body Evolution



Lower Body Evolution

Roughing

Operation: Adaptive Clearing

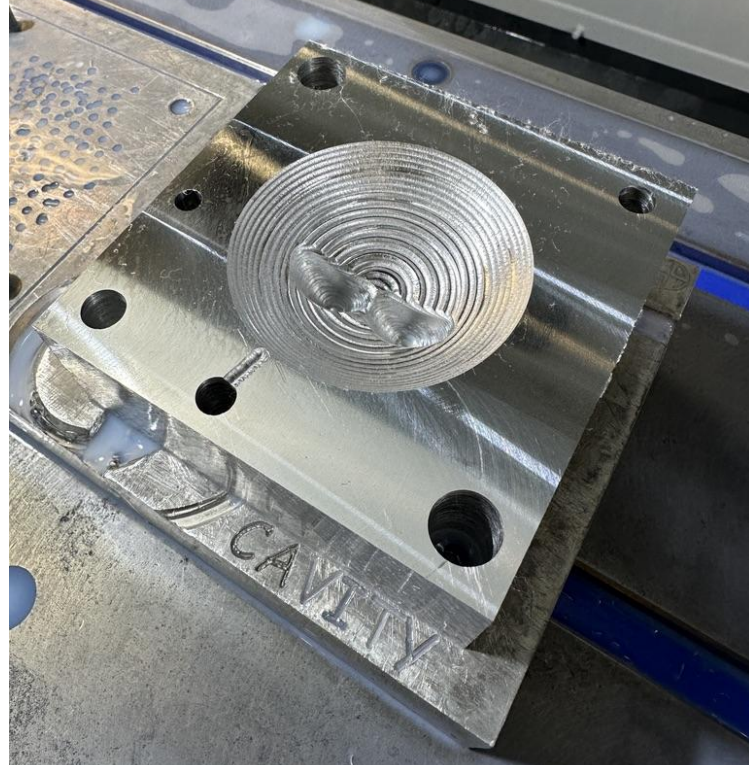
Tool: 1/4" ball mill

Finishing

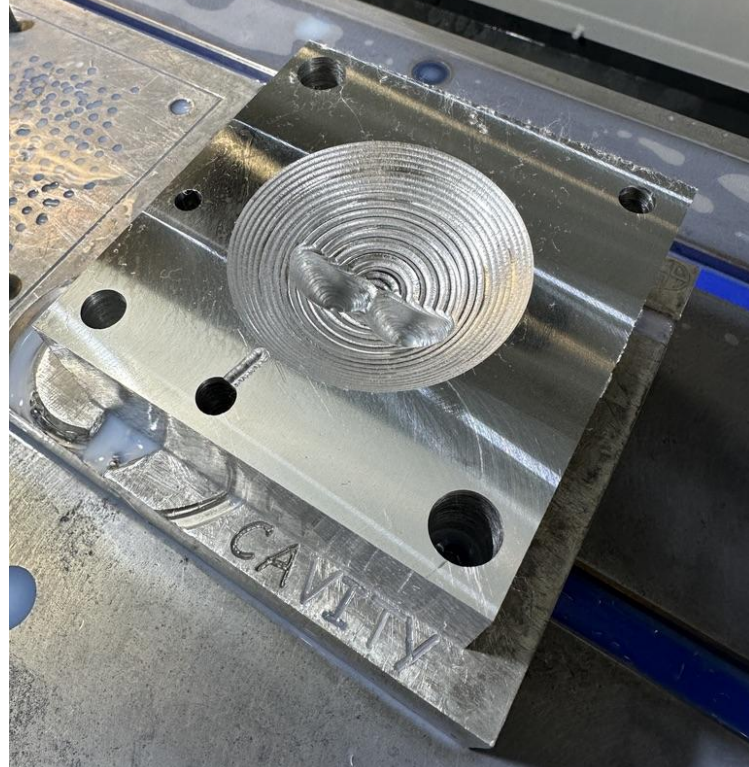
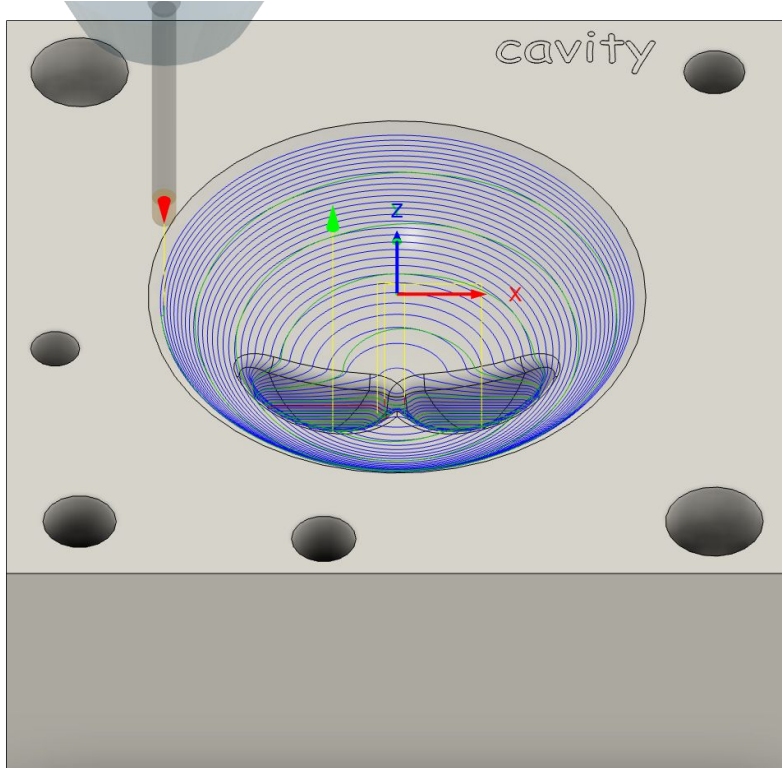
Operation: Contour

Tool: 1/8" ball mill

Cusp Height: 0.001"



Lower Body Evolution



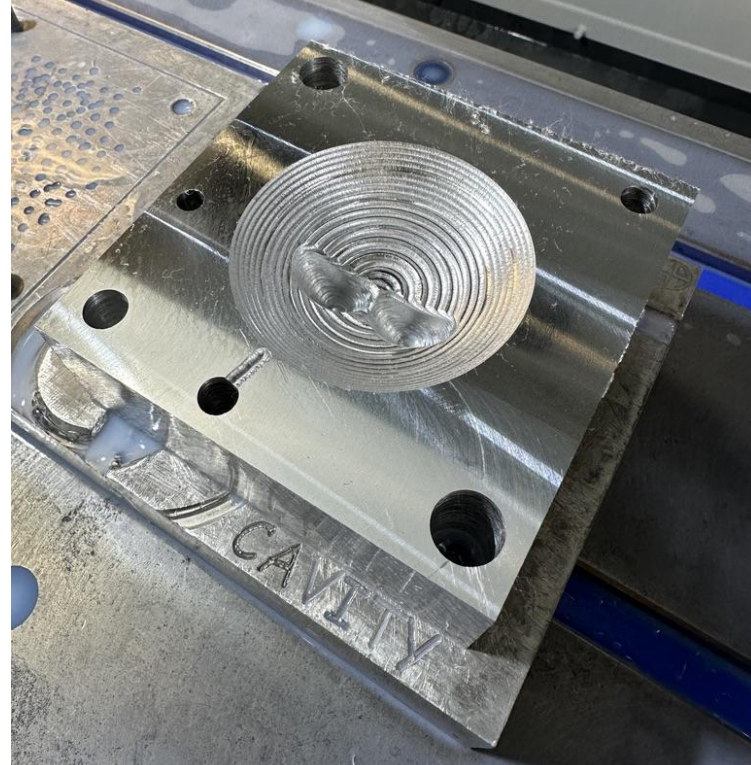
Lower Body Evolution

Attempted Fix #1

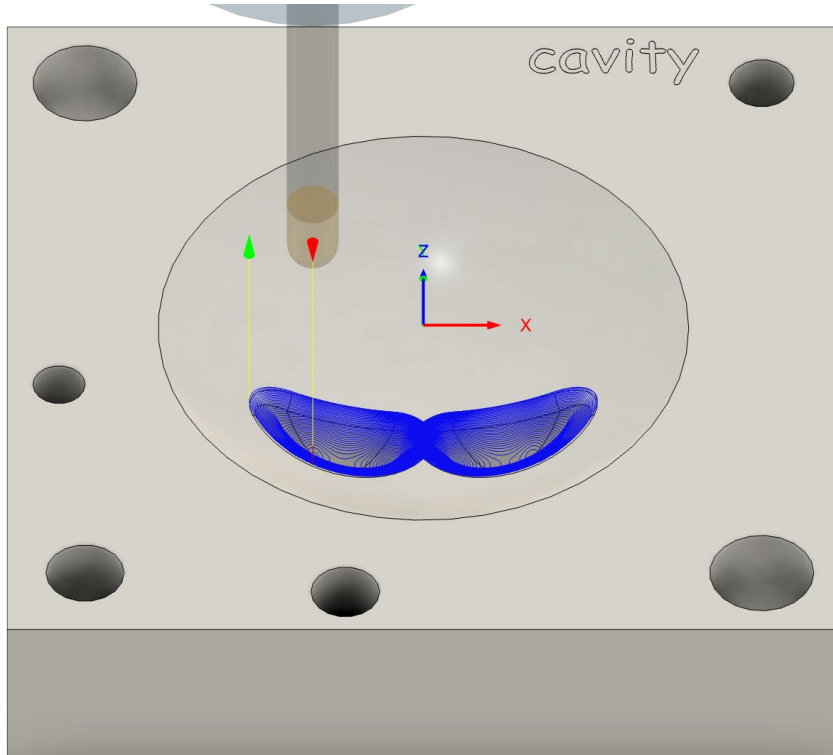
Second contour pass with 3/16" flat end mill in region of poor quality

Attempted Fix #2

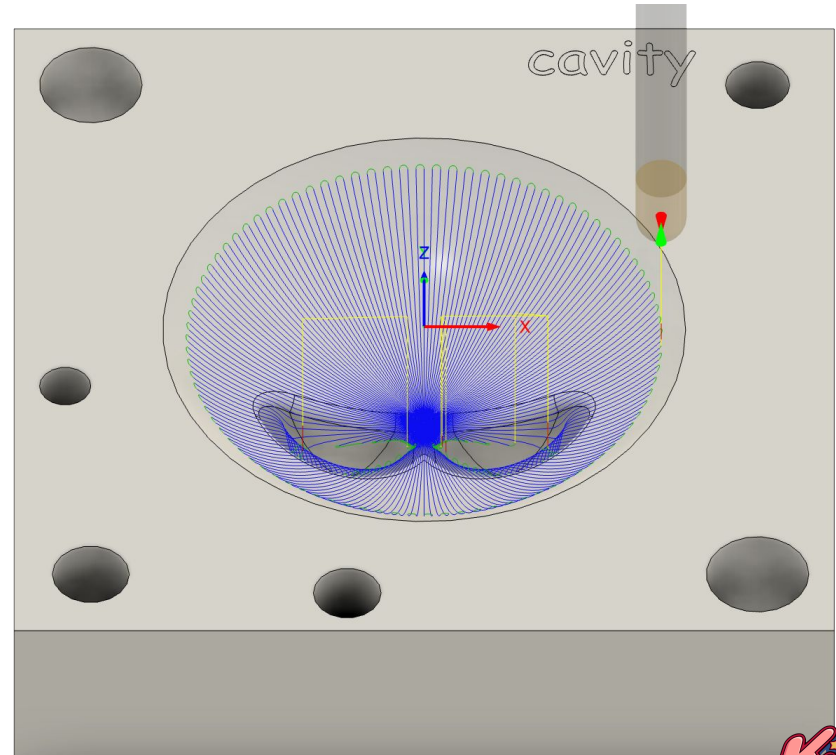
Redo using radially symmetric & freeform finishing operations



Lower Body Evolution

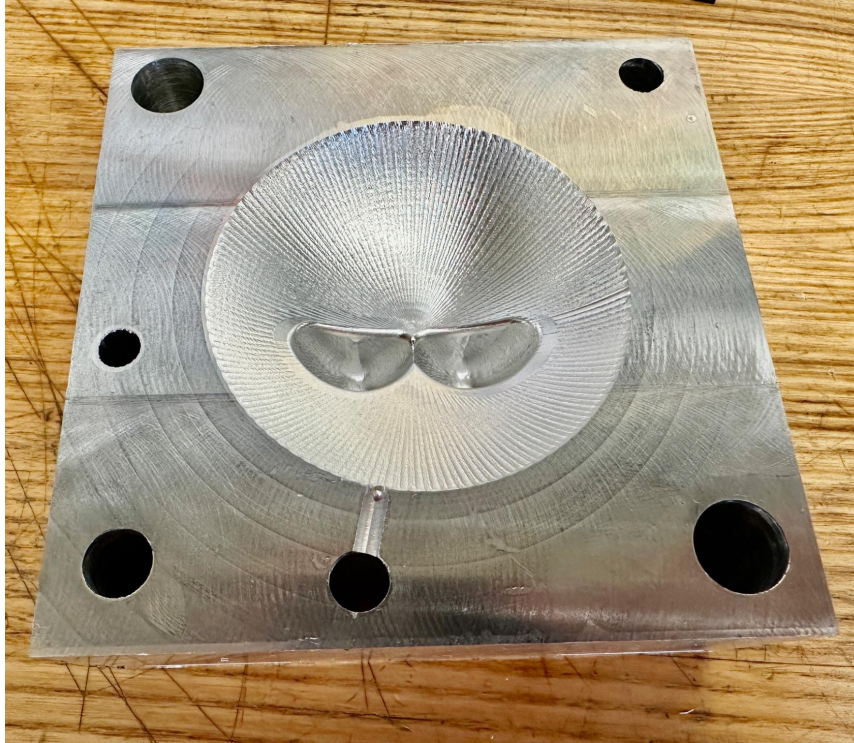


Morphed Spiral | 1/4" ball mill | 0.0001 in cusp height



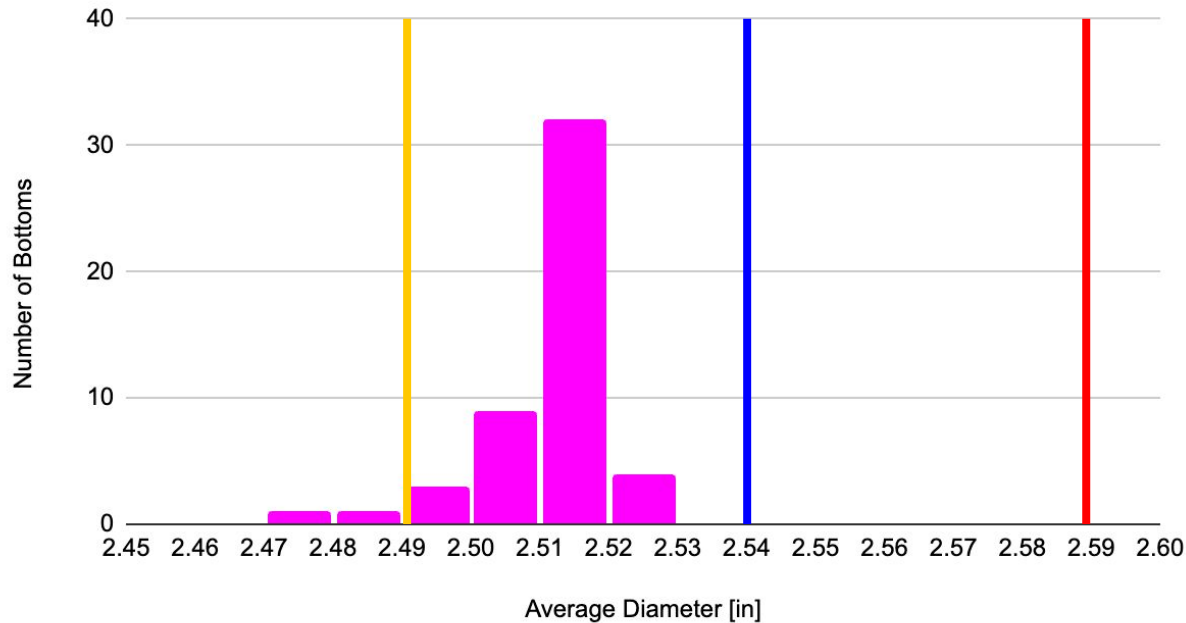
Radial | 2 deg step | 1/4" ball mill

Lower Body Evolution



Distributions - Lower body

Kir-B Lower Body Press Fit Diameter



$$\mu = 2.508984$$

$$\sigma = 0.008$$

$$n = 50$$

CAD SPEC: 2.54" +/- 0.05

USL = 2.59

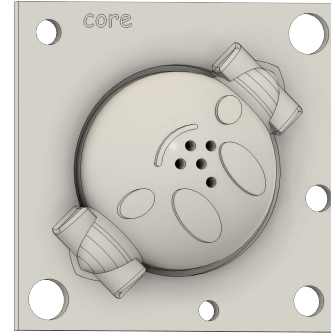
LSL = 2.49

A cartoon illustration of Kirby in a green hat and pink body, standing in a brown, sandy environment. In the background, there are purple, rocky structures. A white circular callout points to Kirby's upper body, containing the text "Kirby's Upper Body Mold".

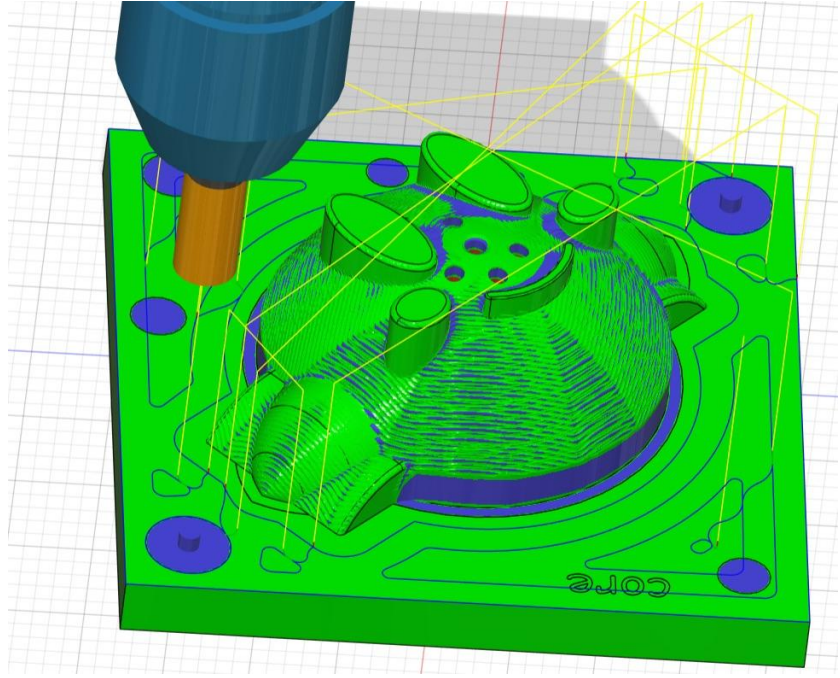
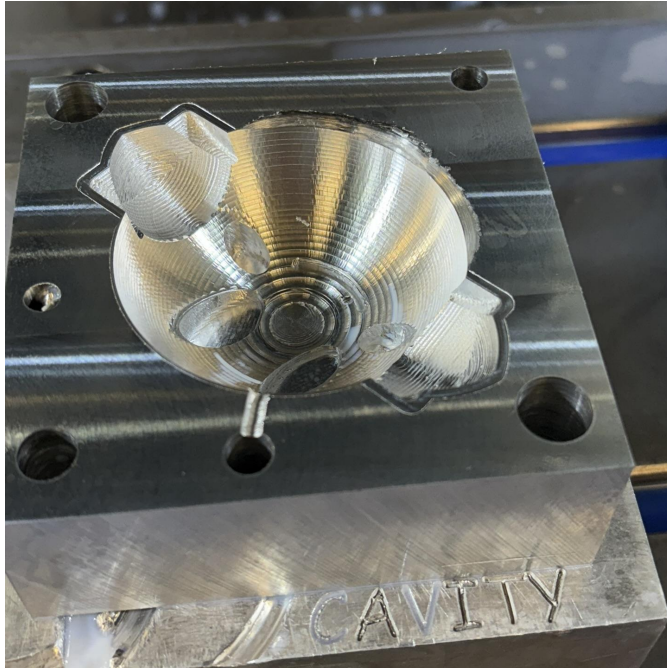
Kirby's
Upper Body
Mold

Source: Reddit user [u/SageeScorpion](#)

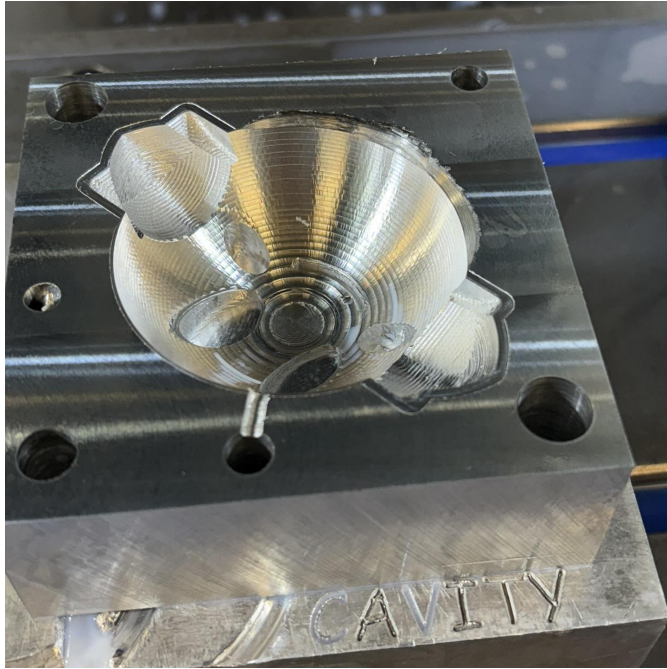
Upper Body Evolution



Upper Body Evolution



Upper Body Evolution

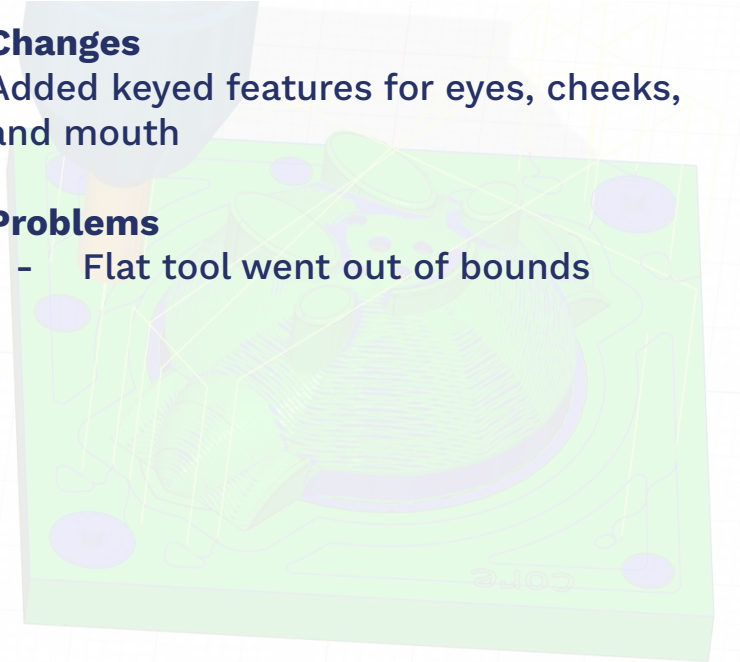


Changes

Added keyed features for eyes, cheeks, and mouth

Problems

- Flat tool went out of bounds



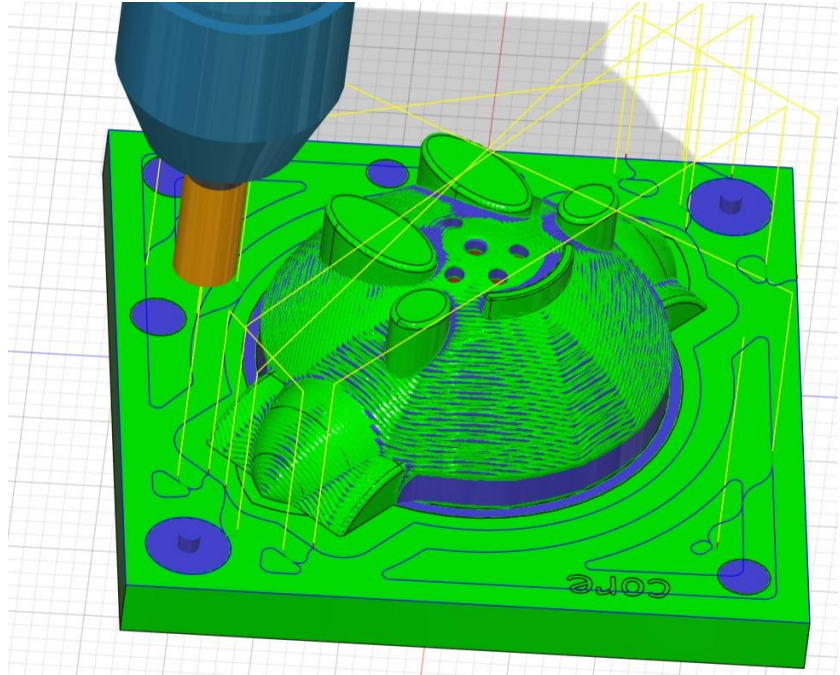
Upper Body Evolution

Problems

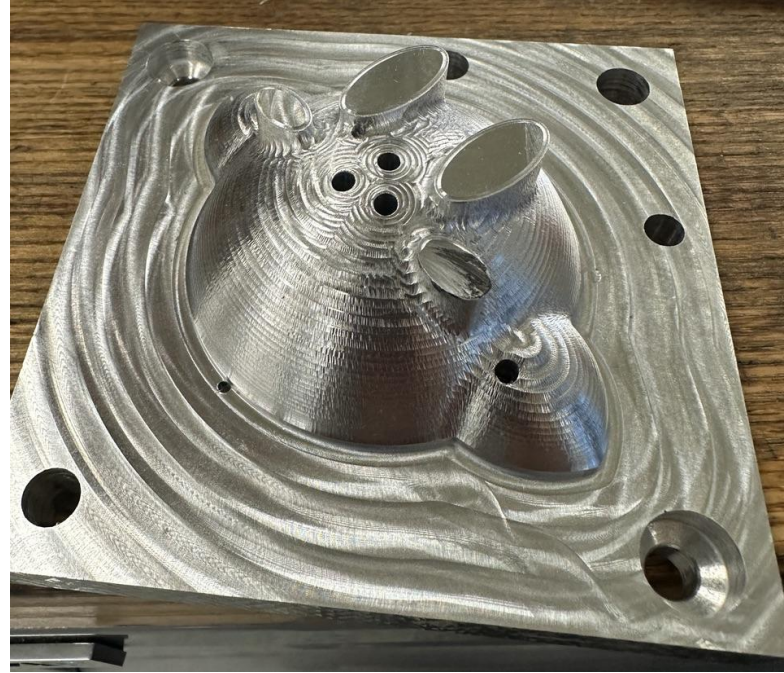
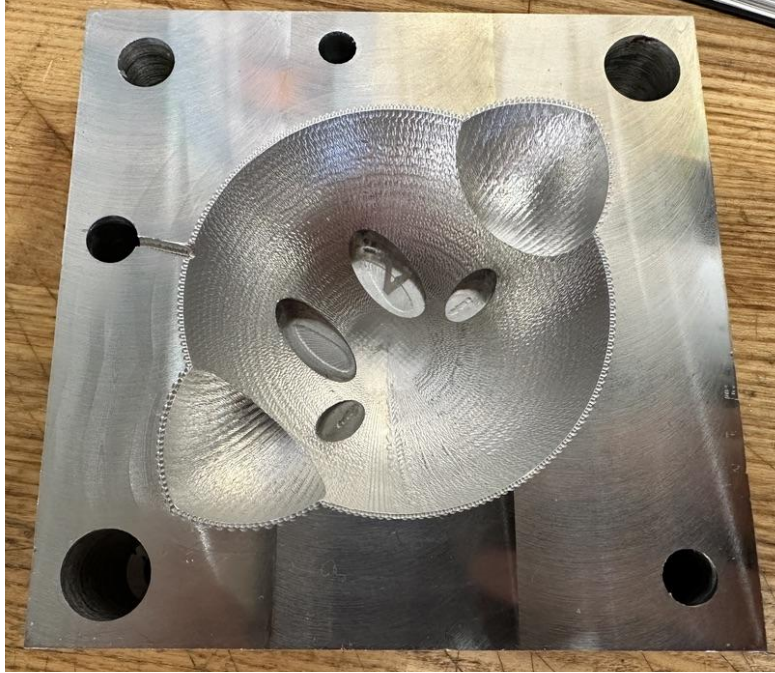
- Fusion has difficulty simulating toolpaths for facial features
- Small diameter tools for contours = finer feed & speed requirements.
- *Stock + Holder* collisions near eyes and cheeks
- Press fit pocket not machinable due to tool holder width

Changes

- Removed mouth
- Eliminated armpit feature
- Edited position of facial features
- Edited eye width and cheek width



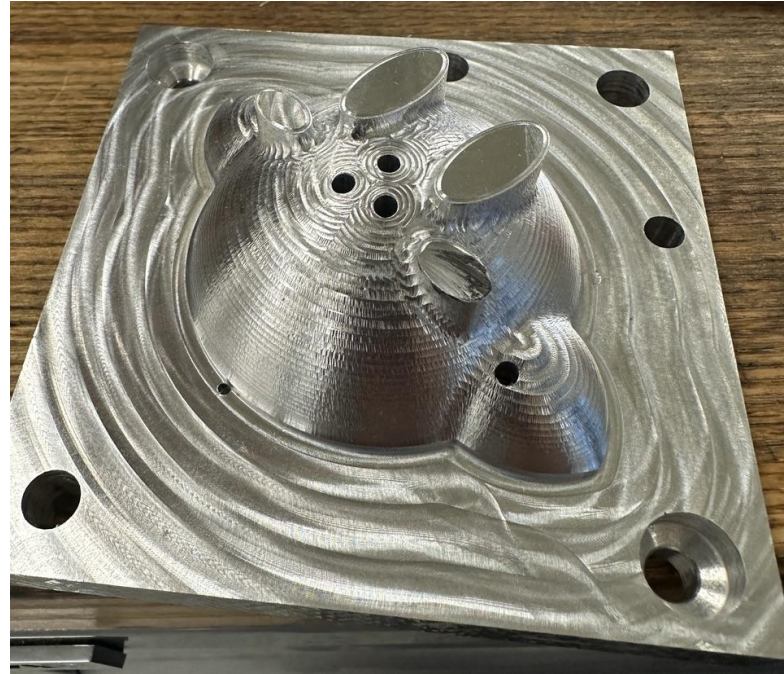
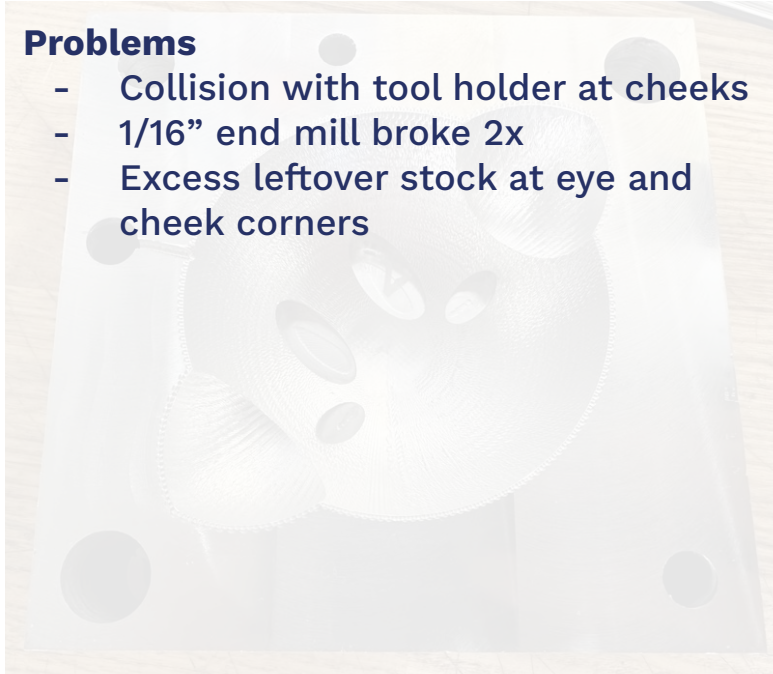
Upper Body Evolution - V2



Upper Body Evolution - V2

Problems

- Collision with tool holder at cheeks
- 1/16" end mill broke 2x
- Excess leftover stock at eye and cheek corners



Upper Body Evolution - V2

Problems

- Collision with tool holder at cheeks
- 1/16" end mill broke 2x
- Excess leftover stock at eye and cheek corners
- Outer press fit surface tapered
- Discovered incorrect radius in CAD (0.030" difference)

Changes

- Filed down interference points at cheek and eye features
- Used 2° taper end mill to make press fit pocket



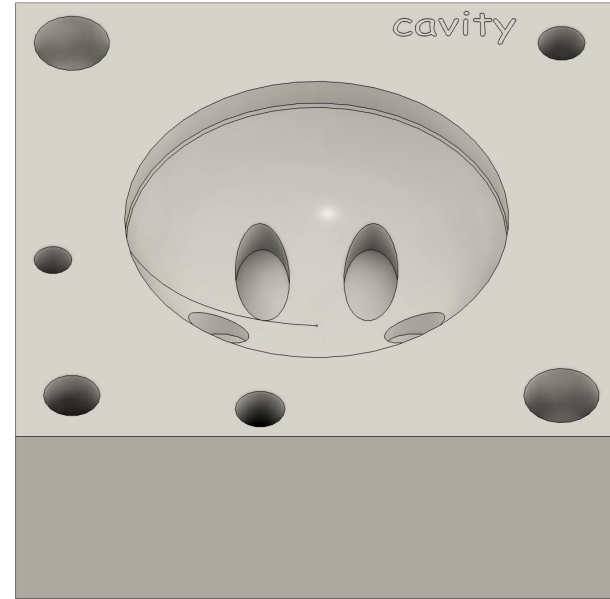
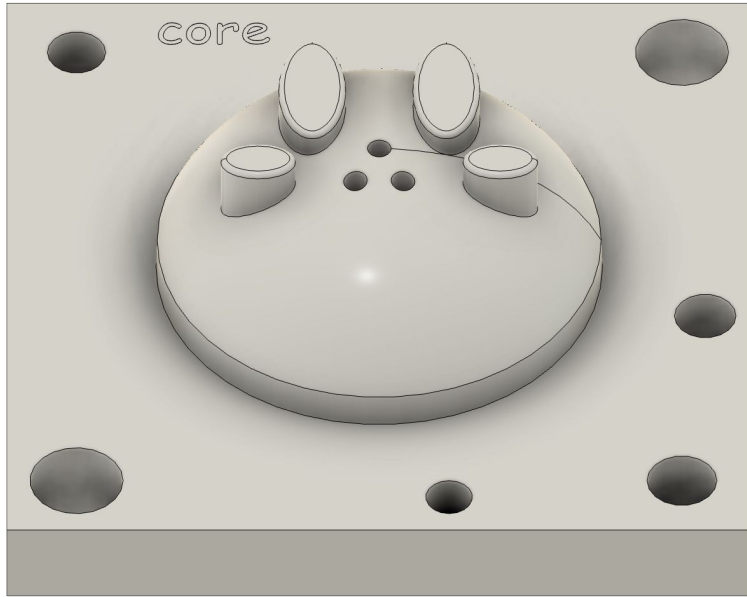
Upper Body Evolution - Armageddon (V3)



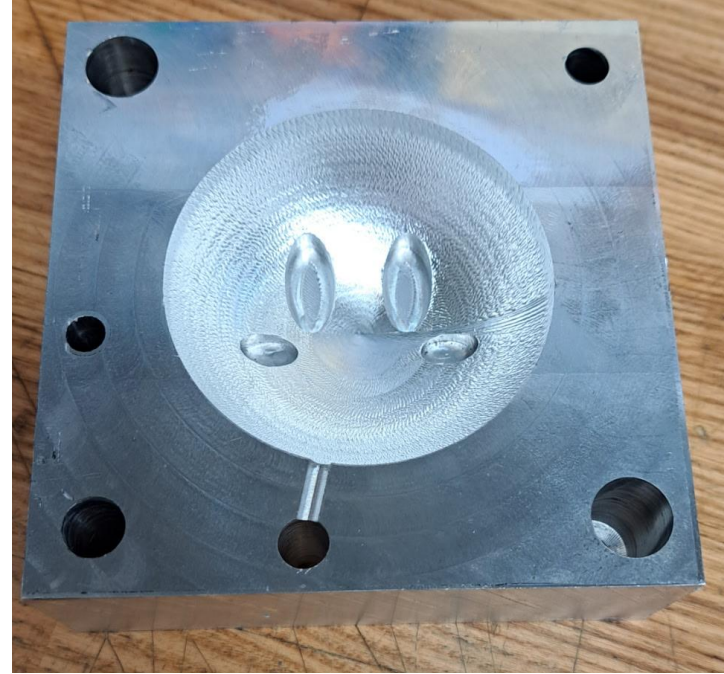
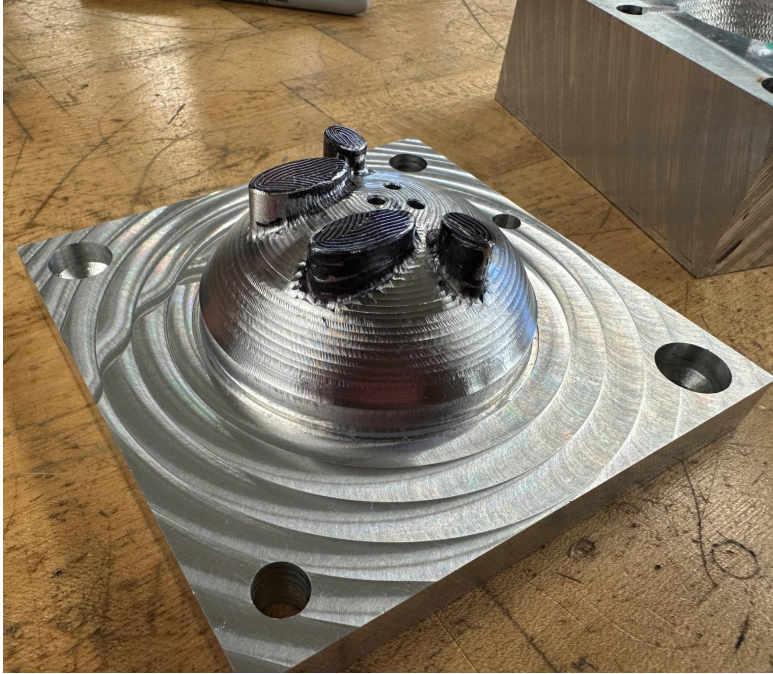
Problems

- Arms caught on floatie
- Press fit worse than on lower body
- Press fit surface still tapered

Upper Body Evolution - Armageddon (V3)



Upper Body Evolution - Armageddon (V3)



Upper Body Evolution - Armageddon (V3)

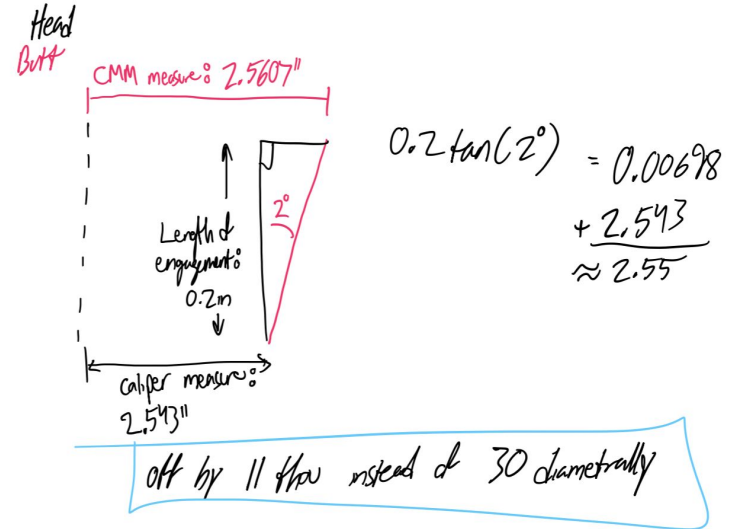
CMM Measurements

Current Values

Date: 1/12/2024

Body, mold diameter:	2.5307	[in]
Ring (upper), mold diameter:	2.5925	[in]
Ring (lower), mold diameter:	2.5607	[in]
Upper-Lower Diff.	0.0318	

Head Mold V2



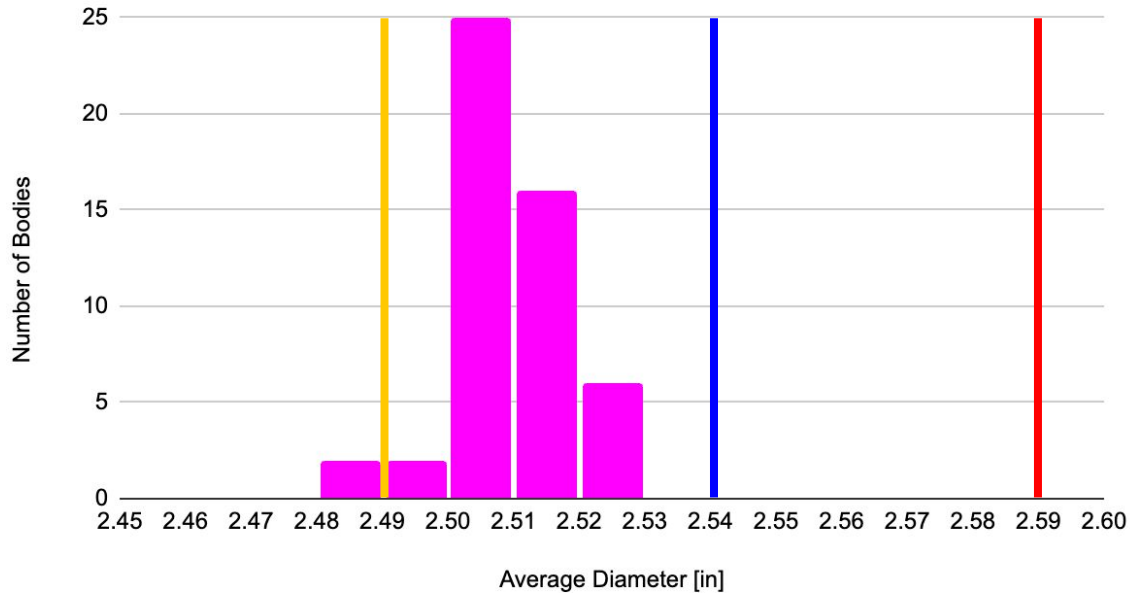
Head Mold V3

Upper Body Evolution - Armageddon (V3)



Distributions - Upper body

Kir-B Upper Body Press Fit Diameter



$$\mu = 2.5121$$

$$\sigma = 0.009$$

$$n = 50$$

CAD SPEC: 2.54" +/- 0.05

USL = 2.59

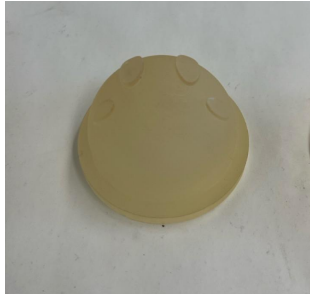
LSL = 2.49

Distributions - Interference

Dimension	Mean	Standard Deviation
OD upper body	2.5121	0.009
OD lower body	2.508984	0.008
ID of floatie	2.4965	0.015
Interference upper	0.0156	0.0175
Interference lower	0.012	0.017

Thermoform

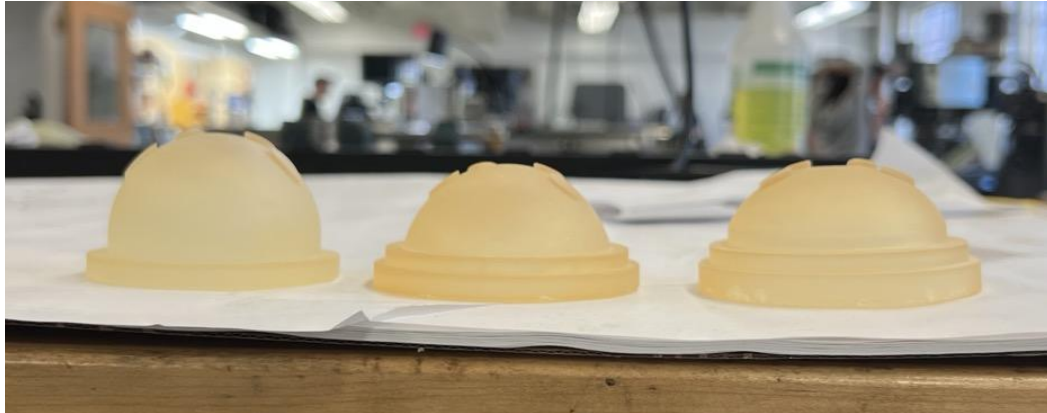
v1



v2



v3



Thermoform

1st TRIAL
(no drill
holes)

SETTINGS
HEATING: 11s
FORMING: 9s



SETTINGS
HEATING: 15s
FORMING: 15s

SETTINGS
HEATING: 20s
FORMING: 15s

OTHER SETTINGS WERE SET CONSTANT

PLATEN DELAY: 3.0s
PLATEN RETURN: 2.0s
EJECT: .0s

VACUUM DELAY: 1s
FAN DELAY: .0s
PRE HEAT: .0s

Press Fit Engineering Analysis

- present the analysis documented in the Manufacturing Spreadsheet, preparing slides that highlight each key decision making step (all tabs highlighted in orange)
 - instructors and LAs will have the spreadsheet open for reference
- explain the full thread of the press-fit features of your yo-yo, connecting the decisions made during the design phase to the results of the manufacturing phase
 - how well did you deliver on your own specifications?
 - how do your body and ring distributions characterize the success of your manufacture processes?
 - using engineering reasoning, why are there differences between your specifications and measured parts?
 - what did you learn along the way?
- students will be graded on:
 - the completeness of the spreadsheet parameters up to this point
 - the soundness of the engineering reasoning used to make key decisions about press fit parameters
 - consistency between the manufacturing spreadsheet, CAD, and the message conveyed in the presentation

Press Fit & Shrinkage Analysis



Manufacturing Spreadsheet

Machined Mold

- Ensuring upper and lower ring have the same critical dimension
 - Allows for finalizing press fit iteratively while modifying the easiest part to manufacture

Injection Molding

- Determining experimental parameters
 - Documenting every change made to injection molding parameters
 - Photographing changes made
 - Discussing with lab staff when challenges were encountered

Press Fit Decision Making

1. Careful calculations and shrinkage analysis
2. Testing phase
3. Recalculating, design changes, remaking updated molds
 - a. Important: only changes on floatie (body) molds
4. Press fit finalized

Specifications:

	Floatie	Lower Body	Upper Body	Interference
CAD	2.5 in	2.5042 in	2.5042 in	0.0042 in
INJECTION MOLDING	2.5046 in	2.5121 in	2.5077 in	0.0031 in 0.0075 in
ERROR	0.18%	0.32%	0.14%	26.2% * 78.6% *

Press Fit Specifications

Specifications

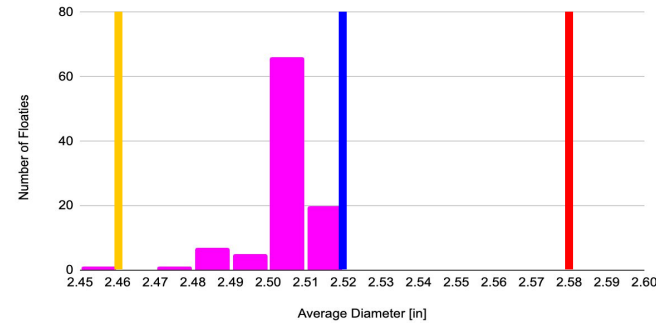
- * variability introduced through:
 - Different people gathering measurements
 - Taking measurements while pieces warm
- * engineering reasoning
 - Machining tolerances and discrepancies
 - Shrinkage

Floatie			Butt			Head		
Body Average:	2.504596667	[in]	Ring Average:	2.512073333	[in]	Ring Average:	2.507661749	[in]
Body Std Dev:	0.01465908895	[in]	Ring Std Dev:	0.008684778791	[in]	Ring Std Dev:	0.008208960144	[in]

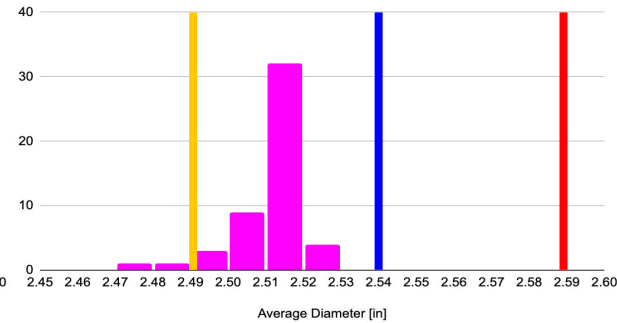
Success: The pieces fit together!

Press Fit Lessons Learned

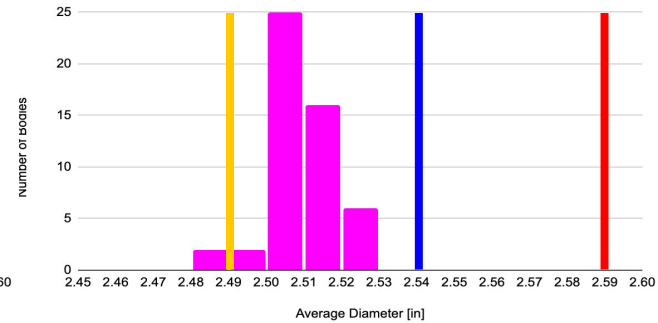
Kir-B Floatie Press Fit Diameters



B Lower Body Press Fit Diameter



Kir-B Upper Body Press Fit Diameter



- 1) Double checking everything
- 2) Nothing goes right the first time
- 3) Rapidly iterable designs
 - a) Press fit corrected iteratively

Organization and Planning

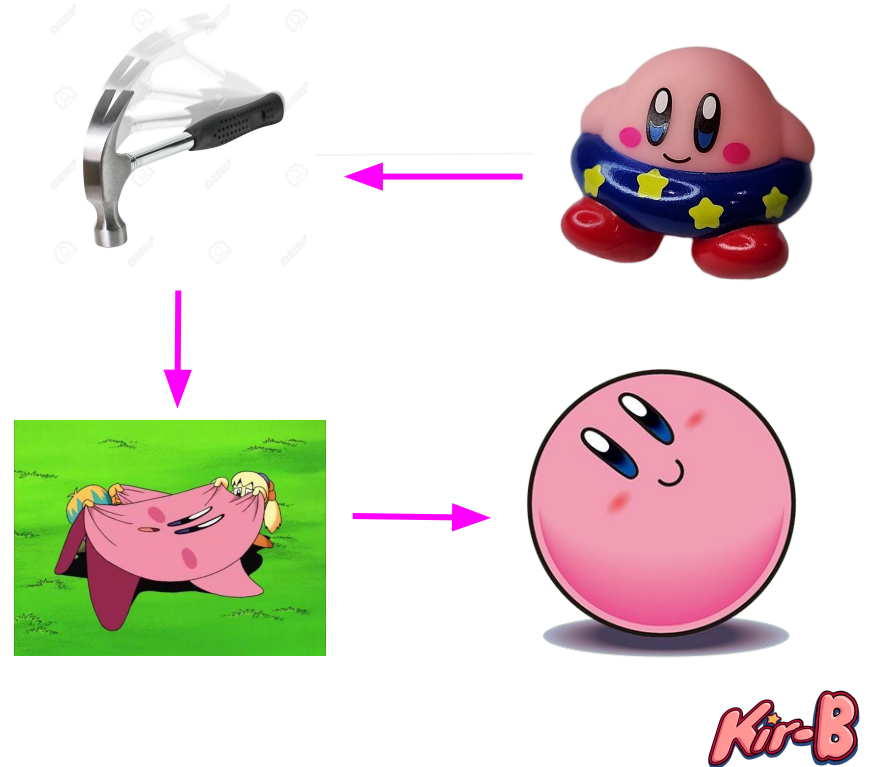
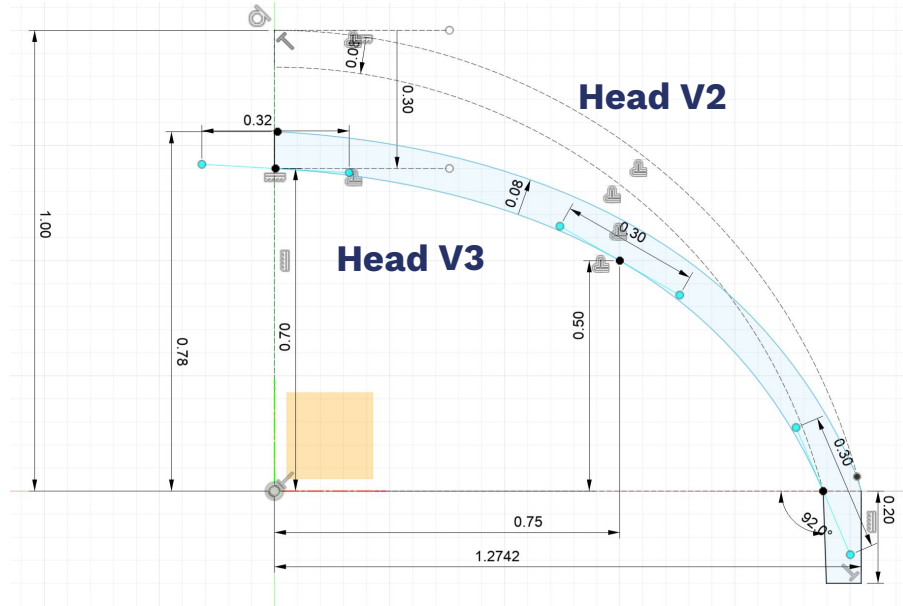
- did your team utilize your backup plan protocol?
 - what was your thinking for managing risk and scaling back your designs?
 - how did your team manage yourselves in terms of getting all the requirements completed?
 - where there any interesting strategic decisions that had to be made to meet the deliverables?
- what is left to do to meet the final 50 yo-yo's lab deliverable?
- students will be graded on
 - demonstration of engineering reasoning applied to strategic decision making throughout the manufacturing portion of the yo-yo project
 - overall clear messaging, effective presentation, and coherent reasoning presented throughout the Manufacturing Reviews

Organization & Planning



Design Contingency Plan Used

- **Make Kirby Smooth**
 - Bye bye arms
 - Reduced Curvature

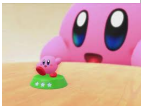


What's left to do?

- Eliminate ejector pin marks on kirby face
- Find the correct pink for kirby halves
- TF 50 Kirby features
- Add star stickers to floatie
- Add eye stickers to face
- Assembly



APRIL 2025

SUN	MON	TUES	WED	THURS	FRI	SAT
		1! 	2	3 	4	5
6	7	8 	9	10 ✗ 	11	12
13	14	15! 	16	17 ✓ 	18	19
20	21 	22! 	23	24 	25 	26
27	28	29 	30			

Team Goals:



Have all IM Parts Fabricated (4/10) ✗ - *did not happen*
Finalize all ring and body molds/solve press fit (4/17) ✓
Assemble one complete Yo-Yo (4/25)

Meetings



Lab (T 2-5PM)



Team Meeting
(7-9PM)

Deliverables



Ring & Body
Molds



IM Parts (Insert
Molded Nuts)



Manufacturing
Review

Holidays



Patriot's Day



MAY 2025

SUN	MON	TUES	WED	THURS	FRI	SAT
				1	2 	3
4	5	6! 	7 	8 	9	10
11	12! 	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Team Goals:



- Assemble 50 Yo-Yos (5/2)
- Finalize final presentation (5/7)

Meetings



Team Meeting
(7-9PM)

Deliverables



50 Yo-Yos (EOL)



Final Presentation
(12:30-2:00 pm)

